MILITARY SPECIFICATION

CHARGE, PROPELLING, XM121 FOR CARTRIDGE, 105-MM, XM546 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of propellant charge designated as Charge, Propelling, 105-MM, XM121, for Cartridge, 105-MM, XM546.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance, Terms and Definitions.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

20-4-618 — Marking Diagram and Sealing of Steel Drums for Shipment of Propellant and Bagged Charges.

76-1-1039 — Drum, Metal, For 75MM and 105MM Howitzer Propelling Charge.

9211657 — Charge, Propelling, 105MM, XM121 For Ctg, APERS, XM546, For Howitzer M101A1, Loading Assembly.

FSC 1315

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PUBLICATIONS

ARMY

IEL–9211657 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling charge assembly. The propelling charge assembly shall comply with all requirements specified on Drawing (dwg.) 9211657 and with all requirements specified in applicable specifications.

3.3 Propelling charge increment weight. The weight of the propelling charge shall not differ by more than the following from that specified in the loading authorization:

Increment number	*Tolerance ounces*
9211660–1	0.02
9211660–2	0.05
9211660–3	0.05

3.4 Workmanship. All components shall be free of dirt, grease and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified critical be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of propelling charges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The examinations and tests listed herein shall be performed on inspection lots submitted in accordance with MIL–STD–105 or MIL–STD–1235, as applicable. The criteria and procedures for the assignment of inspection lot interfix numbers shall be in accordance with Standard MIL–STD–1168. Each lot shall contain:

(a) Single perforated propellant from not more than one lot.

(b) Multiperforated propellant from not more than one lot.

4.2.2 *Examination.* Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect except where 100 percent inspection is specified. Equipmetnt necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 *Body, increment (see dwg. 9211661 covering a detail of dwg. 9211657).*

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.5 percent		
201.	Improper color	Visual	01001
202.	Seam incompletely stitched	Visual	01002

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Categories	Defects	Method of inspection	Code No.
203.	Marking misleading missing or unidentifiable	Visual	01008
204.	Evidence of poor workmanship (see 3.4)	Visual	01004

4.2.2.2 *Increment Parts Arrangement (see dwg. 9211660 covering a detail of dwg. 9211657).*

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.0 percent		
201.	Any seam or opening incompletely stitched	Visual	02001
202.	Evidence of poor workmanship (see 3.4)	Visual	02002

4.2.2.3 *Increment Parts Arrangement (see dwg. 9211778 covering a detail of dwg. 9211657).*

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.0 percent		
201.	Any seam opening incompletely stitched	Visual	03001
202.	Evidence of poor workmanship (see 3.4)	Visual	03002

4.2.2.4 *Increment loading assembly (see dwg. 9211658 or 9211659 covering a detail of dwg. 9211657).*

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Assembly damaged to the extent that propellant can escape	Visual	04001
Minor:	AQL 1.0 percent		
201.	Any seam or opening incompletely stitched	Visual	04002
202.	Thread broken	Visual	04003
203.	Evidence of poor workmanship (see 3.4)	Visual	04004

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4.2.2.5 *Charge assembly (see dwg. 2911657).*

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL .40 percent		
101.	Assembly damaged to the extent that propellant can escape	Visual	05001
Minor:	AQL 1.5 percent		
201.	Increments insecurely tied at junctions	Visual	05002
202.	Ties loose, broken or insecurely tied	Visual	05003
203.	Marking missing, misleading or unidentifiable	Visual	05004
204.	Evidence of poor workmanship (see 3.4)	Visual	05005

4.2.2.6 *Drum, prior to filling (see dwg. 76–1–1039).*

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Foreign material in drums	Visual	06001
Minor:	None defined		

4.2.2.7 *Drum, sealed (see dwg. 76–1–1039 and 20–4–618).*

Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Gasket missing or improperly positioned	Visual	07001
102.	Locking ring improperly positioned	Visual	07002
Minor:	AQL 1.00 percent		
201.	Car seal missing or improperly positioned	Visual	07003
202.	Marking missing, misleading or unidentifiable	Visual	07004

4.2.3 *Testing.*

4.2.3.1 *Propelling charge increment weight (see 3.3)* Major defect. Each increment shall be weighed and check weighed 100 percent. The check weighing shall be accomplished independently of the original weighing using a different balance from that used for the original weighing and if performed manually, it shall be performed by another operator. Any increment which fails to comply with the applicable requirements shall be classified defective and removed from the lot.

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Increment No. 9211660–1	Code No. 08001.
Increment No. 9211660–2	Code No. 08002.
Increment No. 9211660–3	Code No. 08003.

4.2.3.2 *Air pressure test (see 5.1.2) Major defect — Code No. 09001.* The containers shall be checked 100 percent as specified in 4.3.1. If any container fails to meet the requirement it shall be rejected and removed from the lot. Test equipment shall be in accordance with 4.3.4.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Lists Number IEL–9211657 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Air pressure test.* The container shall be subject to an internal pressure of 2 to 3 pounds per square inch (psi) by a method satisfactory to the contracting officer's representative. A gage (designed to read ¼ psi) or a water monometer shall be assembled in the system. A drop of ¼ psi on the gage or 0.7 inch on the monometer in 1 minute shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level C.* The propelling charge shall be packed in a drum conforming to dwg. 76–1–1039.

5.1.2 *Air pressure test of container.* Immediately prior to packing the containers shall be tested with compressed air, at an internal air pressure of 2 to 3 psi. The containers shall withstand the specified pressure for one minute, minimum, without leakage.

5.2 Marking. The drum shall be marked in accordance with dwg. 20–4–618.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

6.2 *Inspection code numbers.* The five-digit code numbers assigned to the inspec-

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tions herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the number column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as Contractor Design in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications — mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication of procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are

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not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35-mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1315–A462

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION		
ORGANIZATION		CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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